

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

Analyst User Guide & Platform Architecture Manual
Comprehensive Operating Guide for Fundamental Research & Screening

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Target Audience	Coverage Scope	Release Version
Equity Research Analysts, Portfolio Managers	92 Nifty 100 Index Constituents (10-13 Years)	v1.0 Production Release

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Core Platform Purpose

The Nifty 100 Financial Intelligence Platform transforms multi-year fundamental financial filings into actionable quantitative and qualitative intelligence. It eliminates manual data aggregation, provides instant access to 50+ financial ratios, automates institutional-grade tearsheet and sector report generation, and powers an interactive 8-page Streamlit dashboard and REST API.

1. Executive Summary & Objectives

This platform serves as the single source of truth for fundamental equity analysis of Nifty 100 constituents across 10-13 years of historical financial filings.

Key Outcomes & Business Value:

- **Unified Data Warehouse:** 10-table relational SQLite database storing P&L, Balance Sheet, Cash Flow, Sector mappings, Stock prices, Market Cap, and Annual Report links.
- **Automated KPI Engine:** Computation of 50+ ratios per company-year combination, handling division by zero, debt-free companies, negative equity, and turnaround growth anomalies.
- **Multi-Criteria Screener:** 6 institutional-grade preset screens and custom filter builder with sector-relative normalisation.
- **Automated Institutional Reporting:** 92 two-page company tearsheets, 11 sector intelligence PDFs, and master portfolio almanac generated via ReportLab.
- **Interactive Analyst Dashboard:** Multi-page Streamlit web application providing zero-code exploratory power.
- **FastAPI REST Server:** 16 REST endpoints delivering structured JSON data for quantitative pipelines.

Core Design Principle: Every insight must be traceable to a formula, every formula to a source column, every column to a dataset file. No black-box numbers.

2. System Architecture & 7-Layer Platform Design

The platform is structured into a modular 7-layer architecture ensuring robust data integrity, separation of concerns, and clean maintainability.

Layer	Layer Name	Components & Technologies
L1	Raw Ingestion	Excel loader (header=1 support), schema validator (pandas, openpyxl)
L2	ETL & Normalisation	Year standardiser (normalize_year), ticker trimmer, dedup engine
L3	Persistent Storage	SQLite 3.x database (data/nifty100.db) with 10 tables and FK constraints
L4	Analytics Engine	Ratio engine (ratios.py), CAGR engine (cagr.py), Cash flow engine
L5	Intelligence Layer	Peer percentiles (peer.py), Health scorer, KMeans clustering, NLP generator
L6	Reporting Layer	ReportLab PDF generation (tearsheets, sector reports, portfolio summary)
L7	Dashboard & API	Streamlit 8-screen app, FastAPI REST server (16 endpoints, OpenAPI 3.0)

3. Data Engineering, Normalisation & DQ Validation

The ETL pipeline handles real-world reporting quirks across Indian corporate filings.

16 Data Quality Rules (DQ-01 to DQ-16):

Rule ID	Rule Name	Condition / Check	Severity
DQ-01	Company PK Uniqueness	len(companies) == companies.id.nunique()	CRITICAL
DQ-02	Annual PK Uniqueness	No duplicate (company_id, year) in time-series	CRITICAL
DQ-03	FK Integrity	All child company_id exist in companies master	CRITICAL
DQ-04	Balance Sheet Balance	total_assets - total_liabilities / assets < 1%	WARNING
DQ-05	OPM Cross-Check	reported_opm - (op/sales*100) < 1.0%	WARNING
DQ-06	Positive Sales	sales > 0 for non-bank operating entities	WARNING
DQ-07	Year Format	Matches r'^\d{4}-\d{2}\$' after normalisation	CRITICAL
DQ-08	Ticker Format	Length 2-12 characters, uppercase stripped	CRITICAL
DQ-09	Net Cash Check	net_cash_flow - (CFO+CFI+CFF) <= 10 Cr	WARNING
DQ-16	Coverage Check	Each company has >= 5 years of history	WARNING

4. Financial Ratio Engine & 50+ KPI Formulas

All formulas follow rigorous institutional equity research standards:

Category	KPI Name	Exact Formula	Edge Case Handling
Profitability	Net Margin (NPM %)	$\text{net_profit} / \text{sales} * 100$	None if sales == 0
Profitability	Operating Margin	$\text{operating_profit} / \text{sales} * 100$	Validated vs opm_percentage
Returns	Return on Equity	$\text{net_profit} / (\text{equity} + \text{reserves}) * 100$	None if total equity <= 0
Returns	Return on Capital	$\text{EBIT} / (\text{equity} + \text{debt}) * 100$	Exclude Banks/NBFCs from D/E flag
Leverage	Debt to Equity	$\text{borrowings} / (\text{equity} + \text{reserves})$	0 for debt-free companies
Leverage	Interest Coverage	$(\text{op_profit} + \text{other_inc}) / \text{interest}$	None if interest=0 (Debt Free)
Cash Flow	Free Cash Flow	$\text{operating_activity} + \text{investing_activity}$	Flag if negative 3yr in a row
Cash Flow	CFO / PAT Ratio	$\text{operating_activity} / \text{net_profit}$	>1.0 = High Quality; <0.5 = Accrual Risk
Growth	5-Yr Rev / PAT CAGR	$((\text{end} / \text{start}) ** (1/5) - 1) * 100$	Turnaround flag if base < 0

5. Investment Screener & Preset Guide

The platform features 6 pre-built screener templates configured via config/screener_config.yaml:

Preset Name	Threshold Filters Applied	Ranking Metric	Target Focus
Quality Compounder	ROE > 15%, D/E < 1.0, FCF > 0, 5Y Rev CAGR > 10%	Composite Score (desc)	High quality compounding compounders
Value Pick	P/E < 20, P/B < 3.0, D/E < 2.0, Div Yield > 1%	FCF Yield (desc)	Undervalued cash generators
Growth Accelerator	5Y PAT CAGR > 20%, 5Y Rev CAGR > 15%, D/E < 2.0	5Y PAT CAGR (desc)	High compounding growth winners
Dividend Champion	Div Yield > 2%, Div Payout < 80%, FCF > 0	Dividend Yield (desc)	Sustainable high yield income
Debt-Free Blue Chip	D/E = 0, ROE > 12%, Sales > ₹5,000 Cr	ROE (desc)	Zero debt pristine balance sheets
Turnaround Watch	3Y Rev CAGR > 10%, FCF > 0	3Y Rev CAGR (desc)	Operational turnaround plays

Important Rule: The Financials sector (Banks, NBFCs, Insurance) is automatically exempted from the D/E screener threshold to avoid structural misclassification.

6. Composite Health Scoring Model (0-100)

The platform computes an objective Composite Quality Score (0 to 100) combining 4 fundamental pillars:

Pillar	Weight	Underlying Inputs & Individual Weights	Normalisation Method
Profitability	35%	ROE (15%), ROCE (10%), Net Profit Margin (10%)	P10/P90 Winsorisation scaled 0-100
Cash Quality	30%	FCF CAGR (15%), CFO/PAT Ratio (10%), Positive FCF (5%)	Scaled 0-100; positive flag=100
Growth	20%	5-Yr Revenue CAGR (10%), 5-Yr PAT CAGR (10%)	Turnaround flag assigned 0
Leverage	15%	Debt-to-Equity Score (10%), ICR Score (5%)	D/E: 0=100, 0.5=85, 1=70, 2=50, >5=0

Score Interpretation Bands:

- **70 - 100 (Tier 1 - Strong Financial Health):** Superior capital efficiency, robust cash generation, low leverage.
- **50 - 69 (Tier 2 - Moderate / Stable):** Standard large-cap performance with stable operations and moderate debt.
- **Below 50 (Tier 3 - Elevated Risk / Watchlist):** Low margins, negative free cash flows, or heavy debt service burden.

7. Peer Comparison & Radar Analytics

Companies are benchmarked against 11 dedicated peer groups across 20 key fundamental metrics:

Peer Group	Benchmark Constituent	Member Constituents
Private Banks	HDFCBANK	HDFCBANK, ICICIBANK, AXISBANK, KOTAKBANK, INDUSINDBK
Public Banks	SBIN	SBIN, BANKBARODA, CANBK, PNB
IT Services	TCS	TCS, INFY, HCLTECH, TECHM, LTIM
Pharmaceuticals	SUNPHARMA	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, TORNTPHARM
Automobiles	MARUTI	MARUTI, TATAMOTORS, M&M:, BAJAJ-AUTO, EICHERMOT, HEROMOTOCO, TVSMOTOR
Life Insurance	LICI	LICI, HDFCLIFE, SBILIFE, ICICIPRULI
Oil & Gas	RELIANCE	RELIANCE, ONGC, BPCL, IOC, GAIL
Power & Utilities	NTPC	NTPC, POWERGRID, TATAPOWER, ADANIPOWER, NHPC, JSWENERGY, ADANIGREEN
Steel & Metals	TATASTEEL	TATASTEEL, JSWSTEEL, JINDALSTEL, HINDALCO
FMCG	HINDUNILVR	HINDUNILVR, ITC, BRITANNIA, DABUR, NESTLEIND, GODREJCP, TATACONSUM
Consumer Finance	BAJFINANCE	BAJFINANCE, CHOLAFIN, SHRIRAMFIN

8. Streamlit Multi-Page Dashboard User Guide

The Streamlit web application (streamlit run src/dashboard/app.py) provides 8 interactive screens:

Screen Name	Key Interactive Features	Data Exports
01. Home / Overview	Nifty 100 macroeconomic health banner, median KPI tiles, sector distribution donut chart	N/A
02. Company Profile	Search 92 tickers, 6 KPI cards, 10-Yr interactive P&L, Balance sheet, Cash flow charts, pros/cons	Direct Tearsheet PDF Download
03. Financial Screener	6 preset filter dropdowns, 10 sidebar slider filters, sector filters, dynamic results grid	CSV & Excel Download
04. Peer Comparison	Select 11 peer groups, side-by-side KPI comparison table, interactive Plotly radar charts	Excel Peer Report
05. Trend Analysis	10-year sparklines, multi-metric overlay mode (overlay up to 3 metrics), YoY growth tables	CSV Historical Export
06. Sector Analysis	Bubble charts (Revenue vs ROE vs Market Cap), sector median bar charts, constituent lists	Sector PDF Download
07. Capital Allocation	Treemap of 92 companies across 8 cash flow patterns (+ - - etc.), drill-down lists	CSV Allocation Matrix
08. Annual Reports	Access 1,585 annual report links by company and year with validation status badges	Direct Document URLs

9. REST API Specification & Endpoints

The platform exposes 16 high-performance REST API endpoints powered by FastAPI (uvicorn src.api.main:app --port 8000):

Endpoint	Method	Description & Query Parameters
/api/v1/companies	GET	List all 92 companies with metadata and sector (?sector=, ?search=)
/api/v1/companies/{ticker}	GET	Full profile: all master fields, latest KPIs, sector, pros/cons
/api/v1/companies/{ticker}/pl	GET	P&L; statement history (?from_year=, ?to_year=)
/api/v1/companies/{ticker}/bs	GET	Balance sheet statement history (?from_year=, ?to_year=)
/api/v1/companies/{ticker}/cashflow	GET	Cash flow statement history (?from_year=, ?to_year=)
/api/v1/companies/{ticker}/ratios	GET	All pre-computed financial ratios (?year=)
/api/v1/companies/{ticker}/tearsheet	GET	Binary stream download of 2-page tearsheet PDF
/api/v1/screener	GET	Multi-parameter screener (?min_roe=, ?max_de=, ?sector=, etc.)
/api/v1/sectors	GET	List 11 sectors with constituent count and median KPIs
/api/v1/peers/{group_name}	GET	All companies in peer group with percentile ranks
/api/v1/portfolio/stats	GET	P10 to P90 distributions for 10 core KPIs
/api/v1/health	GET	Server health check, database row counts, uptime

10. Quality Assurance, Testing & Operational Runbook

Key Operational Commands (Makefile Quick Reference):

- make load: Executes ETL pipeline to load and validate all 12 datasets into SQLite.
- make ratios: Computes 50+ KPIs, capital allocation matrix, and populates financial_ratios.
- make test: Runs 60+ pytest suite and generates HTML report in reports/pytest_report.html.
- make report: Generates all 92 tearsheets, 11 sector reports, and portfolio summary PDF.
- make dashboard: Launches Streamlit multi-page dashboard on port 8501.
- make api: Launches FastAPI server on port 8000.
- make clean: Cleans up pycache and temporary files.

Acceptance & Verification Summary:

All 23 project deliverables and 20 quality gates have been fully executed, tested, and verified with zero critical defects.